

Decision Trees Lab 0

PIMA indians example

Adapted by EVL, FRC and ASP

2025-03-24

Introductory example

The Pima Indians dataset

The Pima Indian Diabetes data set (`PimaIndiansDiabetes2`) is available in the `mlbench` package.

```
data("PimaIndiansDiabetes2", package = "mlbench")
```

The data contains 768 individuals (female) and 9 clinical variables for predicting the probability of individuals in being diabetes-positive or negative:

- `pregnant`: number of times pregnant
- `glucose`: plasma glucose concentration
- `pressure`: diastolic blood pressure (mm Hg)
- `triceps`: triceps skin fold thickness (mm)
- `insulin`: 2-Hour serum insulin (μ U/ml)
- `mass`: body mass index ($\text{weight in kg} / (\text{height in m})^2$)
- `pedigree`: diabetes pedigree function
- `age`: age (years)
- `diabetes`: class variable

```
dplyr::glimpse(PimaIndiansDiabetes2)
```

```

Rows: 768
Columns: 9
$ pregnant <dbl> 6, 1, 8, 1, 0, 5, 3, 10, 2, 8, 4, 10, 10, 1, 5, 7, 0, 7, 1, 1~
$ glucose <dbl> 148, 85, 183, 89, 137, 116, 78, 115, 197, 125, 110, 168, 139,~
$ pressure <dbl> 72, 66, 64, 66, 40, 74, 50, NA, 70, 96, 92, 74, 80, 60, 72, N~
$ triceps <dbl> 35, 29, NA, 23, 35, NA, 32, NA, 45, NA, NA, NA, NA, 23, 19, N~
$ insulin <dbl> NA, NA, NA, 94, 168, NA, 88, NA, 543, NA, NA, NA, NA, 846, 17~
$ mass <dbl> 33.6, 26.6, 23.3, 28.1, 43.1, 25.6, 31.0, 35.3, 30.5, NA, 37.~
$ pedigree <dbl> 0.627, 0.351, 0.672, 0.167, 2.288, 0.201, 0.248, 0.134, 0.158~
$ age <dbl> 50, 31, 32, 21, 33, 30, 26, 29, 53, 54, 30, 34, 57, 59, 51, 3~
$ diabetes <fct> pos, neg, pos, neg, pos, neg, pos, neg, pos, pos, neg, pos, n~

```

A typical classification/prediction problem is to build a model that can distinguish and predict diabetes using some or all the variables in the dataset.

A quick exploration can be done with the `skimr` package:

```

library(skimr)
skim(PimaIndiansDiabetes2)

```

Table 1: Data summary

Name	PimaIndiansDiabetes2
Number of rows	768
Number of columns	9
Column type frequency:	
factor	1
numeric	8
Group variables	None

Variable type: factor

skim_variable	missing	complete	rate	n	unique	top_counts
diabetes	0	1	FALSE	2	neg: 500, pos: 268	

Variable type: numeric

	skim_variable	n_missing	n_complete	mean	sd	p0	p25	p50	p75	p100	hist
pregnant	0	1.00	3.85	3.37	0.00	1.00	3.00	6.00	17.00		
glucose	5	0.99	121.63	90.54	44.00	9.00	117.00	141.00	199.00		
pressure	35	0.95	72.41	12.38	24.00	6.00	72.00	80.00	122.00		
triceps	227	0.70	29.15	10.48	7.00	22.00	29.00	36.00	99.00		
insulin	374	0.51	155.53	118.78	34.00	7.25	125.00	190.00	846.00		
mass	11	0.99	32.46	6.92	18.20	27.50	32.30	36.60	67.10		
pedigree	0	1.00	0.47	0.33	0.08	0.24	0.37	0.63	2.42		
age	0	1.00	33.24	11.76	21.00	24.00	29.00	41.00	81.00		

Building a classification tree

Start building a simple tree with default parameters

```
library(rpart)
model1 <- rpart(diabetes ~., data = PimaIndiansDiabetes2)
# par(xpd = NA) # otherwise on some devices the text is clipped
```

This builds a model consisting of a series of nested decision rules.

```
print(model1)
```

n= 768

```
node), split, n, loss, yval, (yprob)
      * denotes terminal node
```

```
1) root 768 268 neg (0.65104167 0.34895833)
  2) glucose< 127.5 485 94 neg (0.80618557 0.19381443)
    4) age< 28.5 271 23 neg (0.91512915 0.08487085) *
    5) age>=28.5 214 71 neg (0.66822430 0.33177570)
      10) insulin< 142.5 164 48 neg (0.70731707 0.29268293)
        20) glucose< 96.5 51 4 neg (0.92156863 0.07843137) *
        21) glucose>=96.5 113 44 neg (0.61061947 0.38938053)
```

```

42) mass< 26.35 19 0 neg (1.00000000 0.00000000) *
43) mass>=26.35 94 44 neg (0.53191489 0.46808511)
86) pregnant< 5.5 49 15 neg (0.69387755 0.30612245)
172) age< 34.5 25 2 neg (0.92000000 0.08000000) *
173) age>=34.5 24 11 pos (0.45833333 0.54166667)
346) pressure>=77 10 2 neg (0.80000000 0.20000000) *
347) pressure< 77 14 3 pos (0.21428571 0.78571429) *
87) pregnant>=5.5 45 16 pos (0.35555556 0.64444444) *
11) insulin>=142.5 50 23 neg (0.54000000 0.46000000)
22) age>=56.5 12 1 neg (0.91666667 0.08333333) *
23) age< 56.5 38 16 pos (0.42105263 0.57894737)
46) age>=33.5 29 14 neg (0.51724138 0.48275862)
92) triceps>=27 22 8 neg (0.63636364 0.36363636) *
93) triceps< 27 7 1 pos (0.14285714 0.85714286) *
47) age< 33.5 9 1 pos (0.11111111 0.88888889) *
3) glucose>=127.5 283 109 pos (0.38515901 0.61484099)
6) mass< 29.95 75 24 neg (0.68000000 0.32000000) *
7) mass>=29.95 208 58 pos (0.27884615 0.72115385)
14) glucose< 157.5 116 46 pos (0.39655172 0.60344828)
28) age< 30.5 50 23 neg (0.54000000 0.46000000)
56) pressure>=73 29 10 neg (0.65517241 0.34482759)
112) mass< 41.8 20 4 neg (0.80000000 0.20000000) *
113) mass>=41.8 9 3 pos (0.33333333 0.66666667) *
57) pressure< 73 21 8 pos (0.38095238 0.61904762) *
29) age>=30.5 66 19 pos (0.28787879 0.71212121) *
15) glucose>=157.5 92 12 pos (0.13043478 0.86956522) *

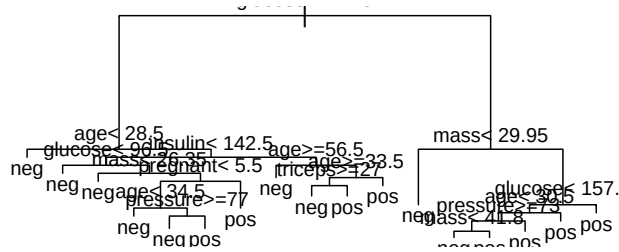
```

The model can be visualized using a tree:

```

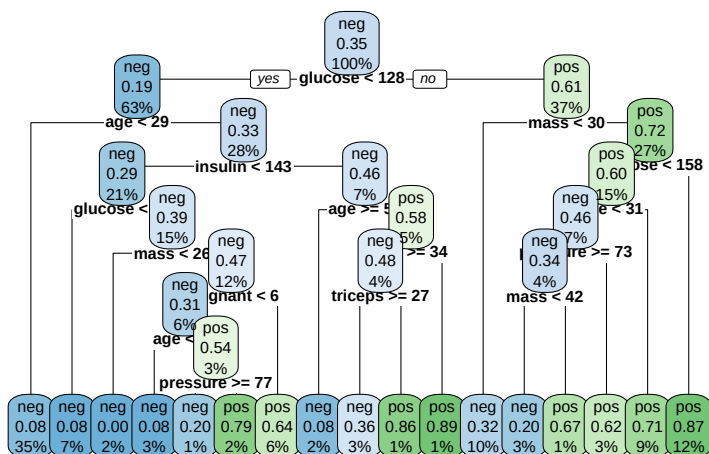
plot(model1)
text(model1, digits = 3, cex=0.8)

```



A nicer plot can be obtained using the `rpart.plot` function from the `rpart.plot` package. This function allows for multiple tunings, but the default values may already yield a nice informative plot.

```
library(rpart.plot)
rpart.plot(model1, cex=.7)
```



```
detach(package:rpart.plot)
```

Assessing model performance

Imagine we know nothing about overfitting.

We may want to check the accuracy of the model on the dataset we have used to build it.

```
predicted.classes<- predict(model1, PimaIndiansDiabetes2, "class")
mean(predicted.classes == PimaIndiansDiabetes2$diabetes)
```

```
[1] 0.8294271
```

A better strategy is to use train dataset to build the model and a test dataset to check how it works.

```
set.seed(123)
ssize <- nrow(PimaIndiansDiabetes2)
propTrain <- 0.8
training.indices <- sample(1:ssize, floor(ssize*propTrain))
train.data <- PimaIndiansDiabetes2[training.indices, ]
test.data <- PimaIndiansDiabetes2[-training.indices, ]
```

Now we build the model on the train data and check its accuracy on the test data.

```
model2 <- rpart(diabetes ~., data = train.data)
predicted.classes.test<- predict(model2, test.data, "class")
mean(predicted.classes.test == test.data$diabetes)
```

```
[1] 0.7272727
```

The accuracy is good, but smaller, as expected.

Making predictions with the model

As an example on how to use the model we want to predict the class of individuals 521 and 562

```
(aSample<- PimaIndiansDiabetes2[c(521,562),])
```

	pregnant	glucose	pressure	triceps	insulin	mass	pedigree	age	diabetes
521	2	68	70	32	66	25.0	0.187	25	neg
562	0	198	66	32	274	41.3	0.502	28	pos

```
predict(model1, aSample, "class")
```

521 562

neg pos

Levels: neg pos

- If we follow individuals 521 and 562 along the tree, we reach the same prediction.
- The tree provides not only a classification but also an explanation.